AI-POWERED INTELLIGENT TALENT ACQUISITION SYSTEM

INDIVIDUAL THESIS

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I. DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. I declare that this thesis is my own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of my knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. I grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this thesis, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

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Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

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II. ABSTRACT

Technical hiring interviews are high-stakes assessments, yet scoring spoken answers consistently is difficult because human judgments vary, speech transcripts are noisy, and surface similarity can reward a fluent answer that is technically incomplete or incorrect. This thesis presents the interview-evaluation component of an AI-Powered Intelligent Talent Acquisition System (project R26-IT-082). The component operates as an independent assessment layer beside CV-to-job matching, resume skill analysis, and early-attrition prediction. It does not replace those modules and it does not make the hiring decision.

The implemented solution uses a containerized microservice architecture. A React and Vite recruiter interface opens a LiveKit interview room. A local faster-whisper speech service transcribes English interview audio. An interview-analysis service reconstructs question-answer pairs, builds or accepts a reference answer, and calls a production multi-task DeBERTa-v3 short-answer grading model. The scorer combines contextual encoder representations, eight lexical coverage features, and three frozen natural-language-inference probabilities. Saved calibration and a five-class decision rule produce a continuous score from 0 to 100 and a rating of Wrong, Poor, Average, Good, or Excellent. A parallel natural-language-inference analyses reports question relevance, concept coverage, and contradiction warnings. Speech-delivery indicators, including filler words, response latency, duration, and word count, are reported separately from the technical score. When the scoring service is unavailable, the result records that failure explicitly and does not present the fallback analysis as the frozen DeBERTa score.

The scoring model was trained on 13,860 leakage-cleaned public automatic short-answer grading examples, validated on 1,685 examples, and tested on 1,705 held-out examples. On that held-out set the model achieved a quadratic weighted kappa of 0.758 and a Pearson correlation of 0.795. Construct validity for technical interviews was examined in a tri-blind pilot of 10 software-engineering items rated independently by three experts on a six-dimension rubric derived from Bloom's Revised Taxonomy. Inter-rater reliability was strong on continuous totals (mean pairwise Pearson r = 0.908) and moderate on five-tier labels (Fleiss' kappa = 0.520). Against human consensus, the calibrated model achieved Pearson r = 0.429 (p = 0.216), Spearman rho = 0.587, quadratic weighted kappa = 0.259, mean absolute error = 9.06, root mean square error = 11.58, and 60.0% exact class agreement. The pilot is below the pre-registered target of 60 to 100 items, so these interview-domain results are preliminary.

The Topic Assessment Form also names certification validation within this member objective; automated authenticity checking of external certificates is treated as future work and is not claimed as an implemented result. The individual contribution covers problem formulation, dataset and feature design, model integration, reference handling, speech analysis, explainable evidence, testing, and the recruiter-facing interview workflow. The final recruitment decision remains with the human recruiter.

Keywords: Automated Interview Scoring, Automatic Short Answer Grading,

DeBERTa, Natural Language Inference, Speech-to-Text, Human Expert Validation.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
ASAG	Automatic Short-Answer Grading
ASR	Automatic Speech Recognition
ATS	Applicant Tracking System
BERT	Bidirectional Encoder Representations from Transformers
DeBERTa	Decoding-enhanced BERT with Disentangled Attention
HR	Human Resources
MAE	Mean Absolute Error
ML	Machine Learning
NLP	Natural Language Processing
NLI	Natural Language Inference
QWK	Quadratic Weighted Kappa
RMSE	Root Mean Square Error
SaaS	Software as a Service
SDG	Sustainable Development Goal
STT	Speech to Text
WebRTC	Web Real-Time Communication
WER	Word Error Rate
XAI	Explainable Artificial Intelligence

1 INTRODUCTION

1.1 Overall Project Context

Recruitment has moved onto digital platforms and applicant tracking systems, yet many of those systems still depend on keyword filters, manual shortlisting, and interviewer judgement. The group project, titled AI-Powered Intelligent Talent Acquisition System and registered as R26-IT-082 under the Centre of Excellence for AI, responds to four related weaknesses: shallow resume screening, opaque job-candidate matching, subjective interview assessment, and the absence of a pre-hire view of early attrition risk. The Topic Assessment Form defines a single platform with four member components that share candidate and vacancy data while keeping their analytical outputs distinct.

Fonseka W.G.K. (IT22248794) is responsible for hiring-risk and early-attrition prediction. Jayantha C.H. (IT22226914) is responsible for intelligent resume parsing and skill-proficiency analysis. Deshan M.A.K. (IT22195616) is responsible for job-candidate matching and explainable recommendations. This thesis covers the fourth component, assigned to Tennakoon T.M.S.S. (IT22197832): AI-based interview evaluation. The Topic Assessment Form also names certification validation within the same member objective. The work reported here designs, implements, and evaluates the interview-scoring pipeline. Automated authenticity checking of external certificates is stated as assigned scope and is treated as future work, because it was not part of the frozen scoring model or the human-validation study.

The shared product, implemented as the Lanka Talent Insights workspace, lets a recruiter manage vacancies, review candidate profiles, compare a CV with a job, schedule an interview, and inspect analytical evidence before making an offer. Each analytical service returns its own result. A strong CV match, a high interview score, and a low attrition signal are three different kinds of evidence. Collapsing them into one automatic hire or reject decision would hide the reason a recruiter should look again.

1.2 Individual Component Context

The interview-evaluation module assesses how a candidate answered the questions that were actually asked. It leaves the CV match score unchanged. After a live interview, the recruiter can inspect, for each detected question, the transcript, the reference answer and its source, a 0 to 100 technical score, a five-class rating, model confidence, concept-level coverage, contradiction warnings, unanswered status, and separate speech-delivery metrics.

The production scorer is a calibrated multi-task DeBERTa-v3 model, identified in the implementation as Final_ASAG_Interview_Scorer_V5. It is an automatic short-answer grading model adapted to technical interview answers. The input at scoring time is the interviewer question, a reference answer, and the candidate response. Demographic attributes, later interview turns, the final hiring decision, and the human validation scores are excluded from that input. This boundary keeps the deployed scorer aligned with information that exists when the interview is being reviewed.

A second branch uses a frozen DeBERTa natural-language-inference model to explain the answer at concept level. That branch can say whether a concept appears to be covered, not mentioned, or contradicted. It is evidence for the recruiter. It is not a second hidden scorer that quietly replaces the calibrated technical score. Speech indicators are likewise kept beside the score so that hesitation, accent, or answer length cannot directly reduce the mark of a technically correct answer.

1.3 Problem Statement

Technical interviews ask candidates to explain concepts, justify design choices, and describe how they would implement a solution. Two candidates can express the same correct idea with different words. A third candidate can repeat the expected keywords and still be incomplete or wrong. Human interviewers can recognise these differences, but their standards vary with experience, fatigue, and the order of candidates. When the answer is spoken, automatic speech recognition adds a further source of error: a misheard technical term can make a correct explanation look incorrect.

The core difficulty is that technical correctness is not the same as textual similarity. A candidate may describe database normalisation using different terminology from the reference answer while still expressing the correct concept. Conversely, a candidate may use several expected keywords but combine them into an incomplete or technically contradictory explanation. Lexical overlap and embedding similarity are attractive because they are simple, yet both can assign a high score to a fluent answer that contradicts the reference. A similarity-only system therefore cannot reliably distinguish a correct but differently worded answer from an incorrect but keyword-rich one. This is the first gap the component must address.

A second difficulty is that interview answers are usually spoken, not typed. Speech recognition introduces errors that do not exist in written text. A misheard technical term, a dropped word, or an incorrectly segmented sentence can change the meaning of an answer. Because the scorer operates on the transcript rather than the audio, any transcription error propagates into the score. The system must therefore be designed so that transcription quality is visible and so that a low-confidence or unanswered item is flagged rather than silently scored.

A third difficulty is trust. Even if a model produces an accurate score, a recruiter will not rely on it unless the reasoning behind the score can be inspected. A single number with no explanation gives the recruiter no basis for agreeing or disagreeing with the assessment. Explainability is therefore not an optional feature but a requirement for practical adoption. The recruiter needs to see which concepts were covered, which were missing, and whether the answer contradicted an expected concept.

Existing applicant tracking systems rarely close these gaps. Many record a video or a transcript and then leave the judgement entirely to the interviewer. Others apply keyword filtering to resumes but do not evaluate spoken technical answers at all. Very few combine speech transcription, contextual scoring, concept-level evidence, and contradiction detection within a single recruitment workflow. The practical problem addressed by this component is therefore the following: how can a recruitment platform turn a spoken technical interview into a question-level technical

assessment, with an interpretable score and inspectable evidence, while keeping the recruiter responsible for the hiring decision?

This problem statement defines the scope of the component. It is not intended to replace the human interviewer, and it is not intended to produce an autonomous hiring decision. It is intended to provide a structured, explainable, and inspectable technical assessment that supports the recruiter's judgement. The remainder of this thesis describes how the interview-evaluation pipeline addresses this problem, how it was evaluated, and what limitations remain.

1.4 Research Gap

Automatic short-answer grading has shown that transformer models can judge free-text answers more effectively than keyword matching [1]-[4]. Those studies are concentrated in education. A classroom item usually has a stable reference answer and a typed student response. A technical interview adds spoken delivery, domain vocabulary, incomplete answers, and questions that may have more than one acceptable explanation.

Interview products and prototypes have explored speech-to-text, conversational agents, facial-emotion recognition, and virtual interviewers [13]-[21]. Systematic reviews of AI interviewing systems report that linguistic content, emotional cues, and behavioral characteristics are seldom analyzed together, and that many systems return a score without an account of why the score was produced [14]. Natural language inference offers one way to separate support from contradiction, which cosine similarity does not do reliably [6], [7], [10]. Local speech recognition, demonstrated at scale by Whisper, makes it possible to transcribe an interview without sending audio to an external transcription API [5].

Four gaps follow for this project. First, technical interview answers need to be judged for correctness, completeness, depth, relevance, and reasoning, rather than for a single similarity value. Second, few published pipelines join local speech transcription, question-answer reconstruction, contextual transformer scoring, lexical coverage, and natural language inference in one recruiter workflow. Third, human validation is often circular when raters are asked to imitate the model's own grade

boundaries. Fourth, a deployed scorer must refuse to invent a model score when the model service is down. This thesis addresses those gaps for English software-engineering interview answers inside the group platform.

1.5 Aim and Objectives

The aim is to design, implement, and evaluate an explainable interview-assessment module that scores a candidate's spoken technical answers against a question and a reference, reports communication evidence separately, and leaves the hiring decision with the recruiter. Specific objectives are as follows.

- Capture live interview audio in a first-party interview room and transcribe it locally with an English speech model prompted for technical vocabulary.
- Reconstruct interviewer questions and the candidate answers that follow them, and mark a question with no candidate response as unanswered.
- Score each answered question with a frozen multi-task DeBERTa-v3 model that combines contextual representations, lexical coverage, and natural-language-inference probabilities, then calibrate the output to a 0 to 100 score and a five-class rating.
- Provide a parallel explanation of relevance, concept coverage, and contradictions, and label job-context references so they are not mistaken for expert-written reference answers.
- Report speech-delivery indicators without allowing them to change the technical score.
- Compare the frozen scorer with human experts on an independent rubric, and compare it with Sentence-BERT, BERTScore, and ROUGE-L baselines on the same human consensus labels.
- Integrate the module with the shared recruitment platform so that recruiters can run analysis from the interview workflow.

Modelling objectives and system objectives are kept apart. The modelling objectives ask whether a calibrated short-answer model can represent answer quality on public grading data and whether those scores show any alignment with independent interview judgements. The system objectives ask whether that score, together with its

evidence, can be shown to a recruiter without concealing model failure or mixing technical quality with speaking style.

1.6 Research Questions or Hypotheses

How would such a system predict the probability of candidate interview performance through analysis of data from the candidate, the position, and the interview process in light of difficulties with class imbalance and incomplete application processes, but at the same time enable the recruiter to determine the performance factors and test their assessment methods without making an automatic hire?

This overarching question is broken down into four research questions, each addressing a distinct part of the problem. The first three are empirical and can be answered with measurable evidence. The fourth is design-oriented and is answered through the safeguards built into the system.

RQ1. Can a calibrated multi-task DeBERTa-v3 model learn useful continuous and ordinal structure on leakage-cleaned public short-answer data?

This question tests whether the model can produce a meaningful graded score rather than only a binary correct-or-incorrect decision. It is evaluated on the held-out public ASAG test set using continuous metrics such as Mean Absolute Error, Pearson correlation, and Spearman correlation, and ordinal metrics such as Quadratic Weighted Kappa. The held-out set is used only once, after all model-development decisions are complete, so the result reflects genuine generalisation rather than tuning. A positive answer to RQ1 establishes that the scoring function has learned useful structure under controlled conditions, but it does not by itself establish validity for spoken technical interviews. This is why RQ1 is evaluated separately from the human pilot.

RQ2. Do human experts agree strongly on continuous interview totals and only moderately on five-tier labels when they use an independent rubric?

This question tests the reliability of the human reference itself before the model is compared against it. If the experts do not agree with each other, then a low model-human correlation cannot be interpreted as a model failure, because there is no stable

human target to align with. The question is evaluated using pairwise Pearson and Spearman correlation for continuous totals and Fleiss' kappa for categorical agreement. The rubric is independent of the model's internal scoring logic, so the experts are not reproducing the model's decision rule. The expected pattern is strong continuous agreement but weaker categorical agreement, because collapsing a continuous total into five bands discards information and introduces boundary effects.

RQ3. On a blinded pilot, does the proposed model show stronger ordinal agreement and exact-class match with human consensus than Sentence-BERT, BERTScore, and ROUGE-L?

This question tests whether the added components of the proposed model, namely the multi-task heads, lexical coverage features, NLI evidence, and calibration, produce a measurable improvement over simpler similarity-based baselines. Sentence-BERT and BERTScore measure semantic similarity, and ROUGE-L measures lexical overlap. These baselines represent the approaches most commonly used when assessing free-text responses. The comparison is evaluated using Quadratic Weighted Kappa and exact-class agreement, because these metrics account for the distance between predicted and true categories and are therefore more informative than raw accuracy in an ordinal grading task. A positive answer would support the claim that combining contextual representations with explicit lexical and NLI features is more useful than similarity alone. Because the pilot contains only ten items, the result is treated as preliminary and is not used to claim statistical superiority.

RQ4. Which safeguards are required so that automated interview evaluation remains a transparent decision-support tool?

This question is design-oriented rather than statistical. It asks what must be built into the system so that the model supports the recruiter without replacing the recruiter's judgement. The safeguards identified in this thesis are local inference so that interview audio remains within the evaluation environment, explicit reference provenance so that recruiters can distinguish expert-written references from job-context fallbacks, separation of technical scoring from communication indicators so

that hesitation or accent cannot reduce a technically correct score, exclusion of candidate demographic metadata from the scorer input, a needs-review route for weak-reference or unanswered items, model-health reporting, and a refusal to fabricate a score when the scorer is unavailable. Each safeguard is verified through the functional test cases reported in Chapter 5. A positive answer to RQ4 requires that all of these safeguards are present and verified, not only that the model produces an accurate score.

Taken together, the four questions cover model validity, human reliability, baseline comparison, and responsible design. RQ1 and RQ3 provide quantitative evidence, RQ2 establishes the reliability of the human reference, and RQ4 defines the conditions under which the system may be used. The remaining chapters address each question in turn and report the corresponding evidence.

1.7 Scope Limitations and Assumptions

The component evaluates English technical interview answers in a software-engineering setting. The preferred reference answer is written by the recruiter or administrator. If no reference is supplied, the system builds a job-context reference from the vacancy skills, description, and the wording of the question, and it marks that reference as a fallback. Job-context references are weaker evidence than expert-written answers and can be routed for human review.

The technical score uses only the question, the reference, and the candidate answer text. Communication metrics are displayed separately. Optional video in the interview room supports the live conversation. Facial emotion, honesty, personality, and employability are not inferred by the scoring model. The human pilot contains 10 items, which is below the pre-registered target of 60 to 100 items required for a powered estimate of agreement. Public short-answer corpora contain typed answers, so the effect of speech-recognition errors on the score has not yet been measured. Results should not be generalized to behavioral interviews, non-English speech, or open design questions that have many valid answers.

The Topic Assessment Form includes certification validation in this member's objective. The evaluated system records interview evidence and does not verify the

cryptographic authenticity of uploaded certificates. That function remains outside the claims of this thesis. The module also assumes that the interviewer and candidate can be distinguished in the transcript, that questions can be detected from question marks and interrogative cues, and that a recruiter will read low-confidence or needs-review items before acting on them.

1.8 Individual Contribution

My individual contribution is the interview-evaluation research and its product path. This includes the problem formulation, the short-answer grading design, the eight lexical features, the fusion of frozen natural-language-inference probabilities with the DeBERTa encoder, the multi-task scoring and calibration recipe used at inference, question-answer reconstruction, reference provenance, speech-metric extraction, the rule that model unavailability must be visible, the blinded human-validation design, the baseline comparison on the pilot items, and the integration of analysis into the recruiter interview screens. Resume parsing, job-candidate matching, and attrition prediction were implemented by the other group members and are described here only to locate this component in the platform.

The work can be grouped into four areas. The first is **research formulation**. This covers the problem statement, the research gap, the four research questions, and the decision to evaluate the model in two separate lines, one on public short-answer data and one on expert-judged interview answers. It also covers the decision to freeze the model before the human pilot so that the pilot acts as a domain-transfer test rather than a tuning exercise.

The second is **model design and scoring**. This includes the eight lexical coverage features, the use of a frozen DeBERTa natural-language-inference model to produce entailment, contradiction, and neutral probabilities, the fusion of these features with the DeBERTa-v3 encoder representation, the three prediction heads for classification, regression, and ordinal scoring, and the calibration recipe that converts the combined output into a 0 to 100 score and a five-class label. It also includes the threshold selection and the decision to keep probability, thresholded class, and displayed risk score as separate outputs.

The third is **interview processing and evidence**. This includes the question-answer reconstruction logic that groups candidate utterances under the preceding interviewer question, the reference provenance rules that distinguish expert-written references from job-context fallbacks, and the speech-metric extraction that produces response latency, answer duration, word count, meaningful-word count, and filler-word frequency and percentage. It also includes the rule that model unavailability must be recorded explicitly rather than replaced by a fabricated score, and the separation of technical scoring from communication indicators so that speaking style cannot reduce a technically correct answer.

The fourth is **validation and product integration**. This includes the blinded human-validation design with three independent experts and a six-dimension rubric, the conversion of human totals into an ordinal category using a scheme independent of the model's thresholds, and the baseline comparison against Sentence-BERT, BERTScore, and ROUGE-L on the pilot items. It also includes the integration of the analysis into the recruiter interview screens, where the transcript, technical score, predicted class, model confidence, concept coverage, contradiction warnings, unanswered-question status, communication indicators, and model-availability status are displayed as separate evidence dimensions rather than collapsed into a single automatic hiring score.

Resume parsing, job-candidate matching, and attrition prediction were implemented by the other group members and are described here only to locate this component in the platform. The interview-evaluation component is decoupled from the matching component and the attrition component, but it consumes the candidate and job context they provide and contributes its own output to the shared recruiter dashboard. This separation keeps the three evidence dimensions distinct and allows each component to be developed, evaluated, and replaced independently.

1.9 Thesis Structure

This thesis is organized according to the SLIIT individual thesis structure.

Chapter 1 introduces the platform context, individual component, problem, research gap, objectives, scope, and contribution.

Chapter 2 reviews the domain, related research, techniques, and conceptual gap.

Chapter 3 presents the research methodology, datasets, experimental procedure, evaluation measures, ethics, and reproducibility.

Chapter 4 describes product and solution design.

Chapter 5 explains implementation, integration, verification, and operation.

Chapter 6 presents research and product results.

Chapter 7 discusses the findings, contributions, limitations, and implications.

Chapter 8 considers commercialization and sustainability.

Chapter 9 concludes the work and identifies future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

The review covers work that directly informs the interview-scoring component. The sources are research on automatic short-answer grading, transformer language models, natural language inference, speech recognition, human agreement in assessment, and AI interview systems published or preprint between the BERT period and the project year. Commercial interview platforms are included where they show what recruiters already use. Attrition modelling, resume parsing, and job-matching algorithms are outside this review except where they explain why interview evidence must stay separate from the other platform scores.

Each source was read for the task it solves, the signal it uses, whether the score can be explained, and whether a human rater judged the system independently of the model's own rules. That reading produced the gap synthesis in Section 2.5.

2.2 Domain and Theoretical Foundations

A technical interview answer is a short-constructed response. Automatic short-answer grading formalizes this as a comparison between a candidate response and an expected answer, or as a learned mapping from that pair to a score [1]. The response can be correct in meaning while differing in wording, or it can share words with the reference while stating a contradiction. A useful scorer therefore needs semantic agreement, coverage of the required concepts, sensitivity to contradiction, and an overall quality judgement.

DeBERTa represents content and position separately through disentangled attention, and DeBERTa-v3 adds replaced-token detection with gradient-disentangled embedding sharing [6], [7]. Those properties suit technical text, where the role of a term in a sentence matters as much as the term itself. Natural language inference adds a three-way relation between a premise and a hypothesis: entailment, contradiction, or neutral. Used here, the premise is the question together with the

reference answer, and the hypothesis is the candidate answer. The three probabilities are features. They do not by themselves become the published score.

Ordinal assessment is a second foundation. The labels Wrong, Poor, Average, Good, and Excellent are ordered. Predicting Good when the reference label is Excellent is a smaller error than predicting Wrong. Quadratic weighted kappa encodes that distance and is therefore a primary agreement metric, alongside correlation and absolute error on the 0 to 100 scale [1].

Human scores are the external construct. If raters are told to copy the model's thresholds, agreement with the model is partly circular. The validation design in this thesis uses a rubric based on Bloom's Revised Taxonomy and maps the average human total onto academic bands that are independent of the model's saved decision thresholds. Communication quality appears in the human rubric as one dimension worth 10 of 100 marks. In the product, the corresponding speech indicators are shown to the recruiter and are excluded from the automatic technical score, so that delivery cannot silently overwrite technical content.

2.3 Related Research

Condor, Litman, and colleagues studied BERT as an aid to human raters of short answers and argued that automation should assist judgement when responses need interpretation [1]. Sentence-BERT has been used to grade descriptive, comparative, and listing answers by embedding similarity [2], [10]. Ghavidel, Zouaq, and Desmarais compared BERT and XLNet on the short-answer grading task [3]. Kaya and Cicekli reported a hybrid BERT architecture with attention and convolutional components and showed that extra structure on top of a transformer can help [4]. Later transformer studies continue to treat educational short-answer grading as a contextual matching problem [11], [12]. These results support RQ1: a transformer encoder can learn graded answer quality. They do not, by themselves, show that the same model agrees with expert technical-interview ratings.

Speech recognition is the bridge from a live interview to that text model. Radford and colleagues showed that Whisper, trained with large-scale weak supervision, is robust across speakers and recording conditions [5]. Robustness reduces the

transcription barrier. It does not remove it. A single substituted token such as a database or framework name can change an entailment judgement. The methodology therefore treats transcription quality as a limitation to be measured later, by scoring the same answers from manual transcripts and from automatic transcripts.

Interview-specific systems add interaction and behavioural signals. Tighare and colleagues describe Skill Sight as a soft-skill platform that uses facial expression, reaction time, and behaviour [13]. Jagtap and colleagues and Patil and colleagues describe voice agents that transcribe speech and continue a conversation [16], [19]. Yadav and colleagues describe a personalized mock-interview voice agent [17]. Wu and colleagues combine multi-agent question generation with virtual digital humans [18]. Sanjana, Li, and He review AI interviewing systems for technical and professional skills and note the lack of integrated linguistic, emotional, and behavioural analysis, and the lack of feedback that helps a candidate improve [14]. Pandey and colleagues and Varshini and colleagues place interview readiness beside resume and job-alignment functions [15], [20]. Patil, Aher, Bhoite, and Bhosale discuss conversational assessment together with predictive shortlisting [21].

Explainability methods such as local surrogate explanations and Shapley-value attributions are established for tabular and general classifiers [8], [9]. This component does not use those methods on the transformer. The explanation a recruiter sees is the natural-language-inference label for each expected concept, together with the reference source, the class probabilities, and the speech indicators. That choice matches the evidence the model actually computes.

2.4 Existing Systems Products and Techniques

Commercial video-interview products such as HireVue, Spark Hire, and VidCruiter allow a recruiter to record remote answers, schedule sessions, and store candidate records. Coding platforms such as Codility score a programming task automatically. Practice tools such as Google Interview Warmup give a candidate feedback on rehearsal answers. These products solve scheduling, storage, and, in some cases, technical-task scoring. They do not, in the form reviewed for this project, publish a local pipeline that pairs each spoken answer with a reference, scores it with a

calibrated multi-task DeBERTa model, and shows concept-level contradiction warnings beside a separate speech panel.

Three simple text methods are the experimental baselines because they represent common alternatives to a trained grader. Sentence-BERT compares the reference and the candidate answer by cosine similarity of sentence embeddings [10]. BERTScore compares contextual token similarity. ROUGE-L measures the longest common subsequence and therefore rewards lexical overlap. All three can rank some correct paraphrases highly. None of them was designed to penalize a contradiction inside an otherwise similar paragraph, and none of them was calibrated on the project's five-class interview scale. They are included so that RQ3 can be answered with methods a recruiter might otherwise trust.

2.5 Gap Synthesis and Conceptual Framework

The literature supports four design commitments. The scorer should be reference-conditioned and contextual, because wording varies. It should see lexical coverage and natural-language-inference relations, because similarity alone can miss omissions and contradictions. It should be trained and calibrated before any human interview labels are observed, so that the pilot is a transfer test. It should present technical quality and speech delivery as separate evidence, and it should name the reference source.

The conceptual framework used in the rest of the thesis is a five-stage pipeline: live capture, local transcription, question-answer reconstruction, technical scoring with the frozen DeBERTa model, and independent evidence from natural language inference and speech metrics. Human experts enter only in the evaluation chapter. They do not train the production weights. The recruiter enters at the end of the pipeline and remains the decision maker.

3 RESEARCH METHODOLOGY

3.1 Research Design

The study follows a design-science approach. An artefact, the interview-evaluation pipeline, is built to address a practical scoring problem, and the artefact is evaluated in two complementary ways. The first evaluation asks whether the scoring model learns graded structure on public short-answer data that it was trained to predict. The second evaluation asks whether those frozen scores align with independent expert judgements of software-engineering interview answers. The two evaluations are not interchangeable. Success on typed educational answers does not prove validity for spoken technical interviews, and a small interview pilot does not replace the held-out grading test.

The production model weights, calibration values, and class-decision rule were frozen before the human study. Human scores were not used for training, calibration, or model selection. This separation makes the pilot a domain-transfer test. The product evaluation then checks whether the same scoring function can be called from the interview-analysis service, whether question-answer pairing and speech metrics behave as specified, and whether model failure is recorded.

The design-science approach was selected because the research problem is both technical and practical. The technical problem is to produce a reliable and interpretable score for a free-form technical answer. The practical problem is to deliver that score to a recruiter inside a live interview workflow without removing human judgement from the hiring decision. A purely experimental study would not test whether the scoring function can be integrated into a working service, and a purely product-oriented build would not test whether the scoring function is valid against independent expert judgement. The two evaluation lines are therefore used together.

The first evaluation line is **model-centred**. It uses leakage-cleaned public short-answer grading datasets with fixed training, validation, and held-out test partitions. The model is trained only on the training partition, model-development decisions are

made on the validation partition, and the held-out test partition is used once for final reporting. This line measures whether the model learns continuous and ordinal structure under controlled conditions. It does not measure whether the model transfers to noisy spoken interview responses.

The second evaluation line is **human-centred**. It uses software-engineering interview answers scored independently by three domain experts using a six-dimension rubric. The experts do not see the model score or each other's scores. The human consensus is converted into an ordinal category using a grading scheme that is independent of the model's learned decision thresholds. This prevents circular validation, where human raters are asked to reproduce the model's own grading rule. Because the human scores are not used at any earlier stage, the pilot acts as a domain-transfer test rather than a tuning set.

The **product evaluation** is the third line. It checks whether the implemented service behaves as specified. This includes verifying that question-answer pairing correctly groups candidate responses, that speech-delivery metrics such as latency, duration, and filler-word frequency are computed as defined, that the explanation service returns model-derived evidence without recalculating the score, and that a model failure is recorded explicitly rather than replaced by a fabricated score. These checks confirm that the research artefact operates as a deployable decision-support component, not only as an offline model.

Taken together, the three lines give the study a clear separation between model validity, expert alignment, and product behaviour. The two model and human evaluations answer whether the scoring function is meaningful, while the product evaluation answers whether it can be used responsibly inside the recruitment workflow. Keeping these lines separate is important because a strong held-out score alone would not justify deployment, and a working service alone would not justify the accuracy claims.

3.2 Data Participants or Study Materials

Two data lines are used. The model-development line is a leakage-cleaned combination of public automatic short-answer grading sets. It contains 13,860

training examples, 1,685 validation examples, and 1,705 held-out test examples. The held-out test set contains 1,002 SciEntsBank examples, 386 Beetle examples, 258 SAF examples, and 59 Mohler examples. These items are typed short answers with reference answers. They establish RQ1. They are not Sri Lankan interview transcripts, and they are not treated as evidence of local hiring outcomes.

Table 1 : Automatic short-answer grading data used to train and test the scorer

Split	Examples	Role
Training	13,860	Model learning
Validation	1,685	Development decisions
Held-out test	1,705	Final public-data evaluation
Held-out composition	1,002 SciEntsBank; 386 Beetle; 258 SAF; 59 Mohler	Unseen mixture of public ASAG sources

The public ASAG data line was chosen because it provides a controlled setting where reference answers and graded labels are already available. SciEntsBank, Beetle, SAF, and Mohler are established short-answer grading corpora that cover different subject domains, answer lengths, and scoring conventions. Combining them produces a more varied training signal than any single source. The leakage-cleaning step removes duplicates and near-duplicates across the splits so that the held-out test set contains genuinely unseen items. This is important because a model that has already seen similar items during training would produce an inflated test score. The held-out set is used only once, after all model-development decisions are complete. The training and validation partitions are used for parameter learning, calibration, and threshold selection. Human interview scores are not part of this line at any stage.

The interview line is a pilot of 10 software-engineering items. Three domain experts scored each answer independently. The experts did not see the model score, the other experts' scores, or candidate demographic data. The consensus score for an item is the arithmetic mean of the three totals. Because the sample is a pilot, the experts and items are study materials for a preliminary validity check. They are not a representative sample of the Sri Lankan IT labour market.

Each interview item contains an interviewer question, a candidate answer, and a reference answer. The reference answer represents the expected technical content for that question. The three experts scored each answer using a six-dimension rubric covering technical correctness, completeness and concept coverage, domain depth and precision, relevance and focus, reasoning quality, and communication clarity. The rubric is independent of the model's internal scoring logic. The model was not shown the rubric, and the experts were not shown the model's five-class decision rule. This separation is what makes the pilot a domain-transfer test rather than a tuning exercise.

The pilot line is deliberately small and is not presented as a statistically powered evaluation. Its purpose is to test whether the frozen model produces scores that move in the same direction as independent expert judgement. A larger human-validated set of 60 to 100 items is identified in the future work as the next step before any deployment claim can be made. Until that set is scored, the pilot results should be read as preliminary evidence of alignment, not as confirmation of reliability.

The two lines therefore serve different roles. The public ASAG line tests whether the model can learn graded structure under controlled conditions. The interview pilot tests whether that learned structure transfers to expert-judged technical interview answers. Neither line alone is sufficient. Using both, with no shared human labels between them, keeps the evaluation non-circular and separates model validity from domain transfer.

3.3 Experimental or Study Procedure

For each interviewer question the system forms a tuple of question, reference answer, and candidate answer. The DeBERTa-v3 encoder reads the question and reference as the premise and the candidate answer as the continuation. Eight lexical features are computed from token sets: Jaccard similarity between reference and answer, recall of reference tokens in the answer, precision of answer tokens against the reference, coverage of question tokens in the answer, the length ratio of answer to reference, and normalized lengths of the answer, the reference, and the question. A frozen natural-language-inference model, loaded locally from the DeBERTa-v3 cross-

encoder used for entailment, contributes the probabilities of contradiction, neutrality, and entailment. The lexical vector and the three probabilities are concatenated with the encoder representation and passed through a fusion network.

Three heads read the fused representation. A classification head predicts one of five quality classes. A regression head predicts a continuous quality value. An ordinal head models the ordered gaps among the five classes. The training objective combines classification loss, regression loss, ordinal loss, and a consistency term. Focal loss and class weights reduce the effect of class imbalance. At inference the three head outputs are blended with saved weights, and a saved linear calibration maps the blend into the unit interval before scaling to 0 to 100. The class label is then taken from the configured decision rule, either biased logits or thresholds on the fused score. All of these parameters stay fixed during the human study.

Question-answer reconstruction is deterministic. Transcript turns are ordered by time. Interviewer utterances are questions when they contain a question mark or begin with an interrogative cue such as what, why, how, describe, or explain. Candidate turns after a question are joined until the next interviewer turn. A question with an empty candidate side is stored as unanswered and is not given an artificial technical score. The human procedure uses the rubric in Table 3.2. Each expert assigns points on six dimensions. The total is the sum, from 0 to 100. The consensus total is then mapped with the bands in Table 3.3. Those bands are ordinary academic brackets. They are intentionally different from the model's internal cut-offs, so that a rater is not asked to reproduce the model rule.

Table 2 : Six-Dimension Human Evaluation Rubric

Dimension	Maximum	What the Points Represent
Technical correctness	30	Factual and algorithmic accuracy
Completeness and concept coverage	20	Coverage of the points the question requires
Domain depth and precision	15	Terminology, complexity, and trade-offs
Relevance and focus	15	Directness of the answer to the question asked

Dimension	Maximum	What the Points Represent
Reasoning quality	10	Justification of the technical choice
Communication clarity	10	Structure and readability of the explanation
Total	**100**	

Table 3 : Independent Human Grade Bands

Consensus Total	Human Class
0–39	Wrong
40–59	Poor
60–74	Average
75–89	Good
90–100	Excellent

The same consensus labels are used for the three baselines. Sentence-BERT uses cosine similarity between the reference and the candidate answer. BERTScore uses contextual token-level F1, while ROUGE-L uses lexical sequence overlap. No baseline is refit on the ten human scores.

3.4 Measures and Evaluation Metrics

On the public test set, classification quality is summarized by accuracy-style class measures and by quadratic weighted kappa, which penalizes distant class errors more than neighbouring ones. Continuous agreement is summarized by Pearson correlation. The figures reported for RQ1 are the held-out quadratic weighted kappa of 0.758 and the held-out Pearson correlation of 0.795. On the human pilot, inter-rater reliability uses mean pairwise Pearson correlation, mean pairwise Spearman correlation, Fleiss' kappa on the five classes, and mean pairwise quadratic weighted kappa. Model-to-human agreement uses Pearson correlation with its p-value, Spearman correlation, quadratic weighted kappa, mean absolute error, root mean square error, exact class agreement, and macro-F1. The pre-registered interpretation

targets, including a quadratic weighted kappa of at least 0.70 and a Pearson correlation of at least 0.70, require on the order of 60 to 100 items for a stable interval. The pilot is reported against those targets and is not described as having met them. Speech recognition can be measured by word error rate. That metric is defined for a later robustness study. It is not reported as a result of this thesis because the pilot scored answer text rather than paired manual and automatic transcripts.

3.5 Data Analysis

Human consensus is the mean of the three expert totals. Class labels for humans follow Table 3.3. Model class labels follow the frozen decision rule. Agreement statistics are computed on the ten paired totals and on the paired classes. A correlation p-value above 0.05 is reported as a failure to reject the null hypothesis of no linear association at that sample size. It is not rewritten as a significant result. Baseline comparison uses the same human totals. A method can have a smaller absolute error and still have weaker class agreement, or a higher rank correlation and a very large absolute error if its raw similarity was not placed on the 0 to 100 human scale. Chapter 6 therefore reports the full metric row for each method and interprets ordinal agreement and exact class match separately from raw error.

3.6 Ethics Privacy Security and Responsible AI

Interview audio, transcripts, and scores are personal data. The interview room is a LiveKit session, and transcription is performed by a local Whisper model so that audio need not be sent to an external speech API for the scoring path. Access to analysis results is limited to authenticated users in the recruitment workspace. Transcript entries can be corrected by an authorized participant before analysis, which gives a human a chance to repair an obvious recognition error.

The scorer does not receive age, gender, or other demographic attributes. It does not infer emotion, honesty, or personality from the face or the voice. Communication metrics are visible, and they are excluded from the technical score so that accent, pause, or answer length is not silently treated as a lack of knowledge. Low ratings of Wrong, Poor, and Average are marked for review, as are unanswered questions. If

the DeBERTa service cannot be called, the analysis record stores the failure, sets the model status to unavailable or partial, and labels any remaining natural-language-inference output as a fallback source. The interface is a decision-support tool. The recruiter makes the hiring decision. The human pilot used blinded scoring and did not expose model outputs to the raters. Any future collection of live candidate audio for research, beyond operational use inside an consented interview, requires informed consent and institutional approval before the recording is retained as a research dataset. Public grading corpora used for model development are existing educational datasets and are not presented as records of Sri Lankan applicants.

3.7 Reproducibility and Research Artefacts

The artefacts are the training split description, the V5 model configuration and weights, the local natural-language-inference model, the inference code that applies the saved blend and calibration, the interview-analysis service, the speech-to-text service, the human-rubric specification, and the pilot evaluation script with its summary metrics. Model files are loaded from the local service directories. Runtime inference does not download weights from a model hub. The public-data experiment and the human pilot are stored as separate results so that a held-out educational score cannot be quoted as if it were the interview-pilot score. A later camera-ready check should call the same production predict endpoint that the recruiter interface uses, so that research code and serving code are the same function. Until that check is repeated on a larger annotated set, the pilot numbers in Chapter 6 should be read as the output of the frozen evaluation script on the ten expert-scored items.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The module must let a candidate and an interviewer meet in a private room, keep an English transcript, and produce a question-level assessment that a recruiter can audit. Table 4.1 lists the functional requirements that the implementation is required to meet. Table 4.2 lists the non-functional constraints that follow from the research design.

Table 4 : Functional Requirements of the Interview-Evaluation Module

ID	Requirement
FR1	The system shall allow an authorized interviewer and candidate to join a scheduled interview room with audio and optional video.
FR2	The system shall transcribe English speech locally and store speaker-attributed transcript turns.
FR3	The system shall pair each detected interviewer question with the following candidate turns and mark a question with no answer as unanswered.
FR4	The system shall score an answered question using the frozen DeBERTa model when a reference answer is available to the scorer.
FR5	The system shall provide natural-language-inference evidence for relevance, concept coverage, and contradiction, and identify the reference source.
FR6	The system shall report filler-word counts, word counts, response latency, and answer duration separately from the technical score.
FR7	The system shall record model unavailability and shall not label fallback analysis as the DeBERTa score.
FR8	The system shall present the overall interview summary, per-question ratings, and review flags to the recruiter.

The functional requirements define what the module must do. FR1 establishes the interview room as the entry point and requires that both participants are authenticated before they can join. FR2 requires local transcription so that interview audio does not leave the evaluation environment, and requires speaker attribution so that interviewer

questions can be separated from candidate answers. FR3 requires question-answer pairing, because the scorer is reference-conditioned and needs to know which question each answer belongs to. A question with no candidate response is marked as unanswered rather than scored, which avoids assigning a technical score to a missing answer. FR4 requires that scoring uses the frozen DeBERTa model and that a reference answer is present before scoring is attempted. FR5 requires that the natural-language-inference branch produces relevance, concept coverage, and contradiction evidence, and that the reference source is identified so the recruiter can distinguish an expert-written reference from a job-context fallback. FR6 requires that speech metrics are reported separately from the technical score, so that hesitation or speaking style cannot reduce a technically correct answer. FR7 requires that model unavailability is recorded explicitly and that fallback analysis is never presented as a DeBERTa score. FR8 requires that the recruiter sees the overall interview summary, the per-question ratings, and any review flags in a single view.

Table 5 : Non-Functional Requirements

Quality Attribute	Design Constraint
Privacy	Transcription and scoring shall run on local model services. Interview analysis shall be accessible only to authenticated recruitment users.
Integrity	Production model weights and calibration parameters shall be loaded from saved artefacts and shall not be updated through recruiter actions or human pilot scores.
Transparency	Each question shall display its score source, reference source, confidence, concepts, and speech metrics.
Separation of Concerns	The interview score shall not modify the CV match score or the attrition score.
Failure Behaviour	A missing model, an empty answer, or a missing question shall be represented as an explicit state rather than assigned a default high score.
Operability	Each service shall expose a health-check endpoint, and model services shall load their weights at startup.

4.2 Component Architecture Within the Complete Solution

The platform is a set of services behind an API gateway. The interview path uses the gateway, the job service that issues the LiveKit room token, the browser interview room, the speech-to-text service, the interview-analysis service, and the interview-answer model service. Resume extraction, CV matching, resume-strength scoring, and attrition scoring are neighbouring services. They may supply job context, including required skills and the job description, which the interview service uses when it must build a fallback reference. They do not write the interview score.

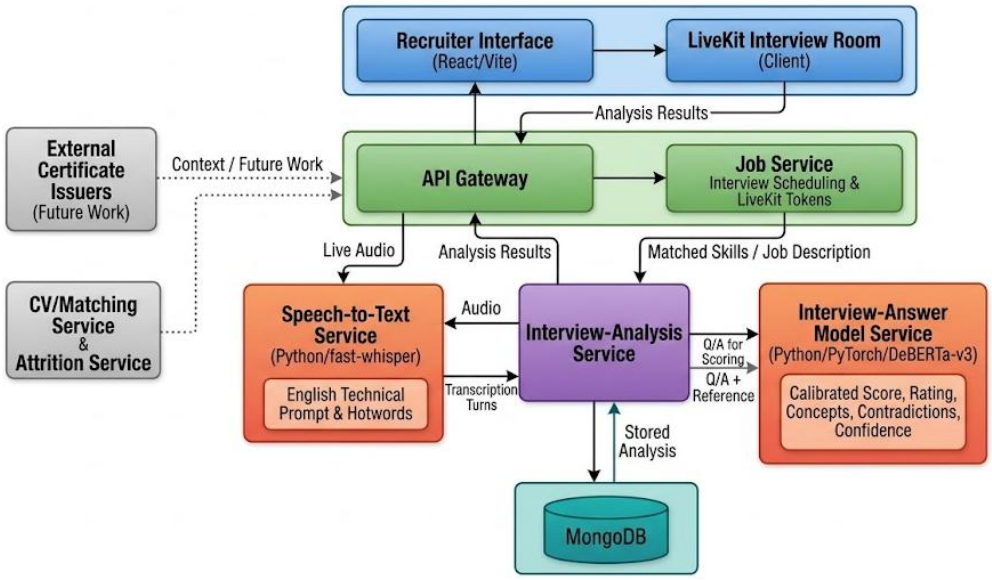


Figure 1 : High-Level Architecture / Component Architecture

This figure illustrates the microservice architecture and data flow of the interview-evaluation component. Interface allows authenticated users to launch and manage sessions through an API Gateway, which coordinates with the Job Service to issue secure LiveKit room tokens. During or after a live interview, audio is transcribed locally by the faster-whisper speech service using domain-specific technical prompts. The Interview-Analysis Service orchestrates question-answer pairing, reference handling, and parallel natural-language-inference (NLI) checks, calling the frozen multi-task DeBERTa-v3 model service to produce calibrated technical scores. Analytical outputs, concept coverage, and separate speech delivery indicators are persisted in MongoDB and displayed back to the recruiter, maintaining strict service boundaries away from resume matching and attrition modules.

Table 6 : Service Boundaries and Responsibilities

Service	Responsibility in This Component
React Interview Room	Provides LiveKit audio and video, a live transcript panel, and an analysis panel for the recruiter.
API Gateway	Provides authenticated access to interview, transcript, and analysis routes.
Job Service	Manages interview scheduling and generates LiveKit access tokens for joining the interview room.
Speech-to-Text Service	Performs local speech transcription using faster-whisper, with an English technical prompt and domain-specific hotwords. Default listening port: 4005.
Interview-Analysis Service	Handles question–answer pairing, reference-plan generation, natural-language-inference (NLI) evidence, speech metrics, scorer orchestration, and persistence of interview analysis. Default listening port: 4006.
Interview-Answer Model Service	Performs frozen V5 DeBERTa scoring using the interview question, reference answer, and candidate answer. Default listening port: 4010.

The processing sequence for one analysis request is as follows. The analysis service loads the interview context, including the transcript and the job. It pairs the transcript. For each pair it judges whether the question is relevant to the job, builds the reference plan, and, when the candidate spoke, calls the model service. It always attaches speech metrics for the pair. It then stores a summary: overall score, question count, answered count, strong-answer count, needs-review count, unanswered count, and how many items were scored by the model. The recruiter reads this record in the interview result view.

4.3 Design Alternatives and Decisions

A separate interview service was chosen so that a change in the scoring model would not require a change to the matching score. Sending audio to a third-party speech API was rejected for the scoring path because the interview should remain inside the deployment. Local faster-whisper with a technical initial prompt and a hotword list was chosen so that terms such as REST, PostgreSQL, Kubernetes, React, and Big-O

are more likely to be preserved. The research configuration described for the study uses an English Whisper model of the small size. The service can load the local model directory configured for the deployment, including a smaller model when that is what has been installed. The prompt and hotword list are part of the design in either case.

A single cosine-similarity score was rejected as the production grader because the literature and the pilot both show that similarity can disagree with expert classes. The production grader is the multi-task DeBERTa model with lexical and inference features. Sentence-BERT, BERTScore, and ROUGE-L remain baselines in the experiment. They are not the score shown to the recruiter. Emotion recognition from the face was considered in the proposal and appears in several related systems [13], [16], [18]. It was not adopted. The technical construct is the quality of the answer content. A face-based emotion score would mix a different construct into the mark and would raise a fairness risk that this study does not claim to have audited. Video remains available for the human conversation. Certificate authenticity checking was part of the member objective on the Topic Assessment Form. It was not implemented as a verification service in the evaluated pipeline. Designing it as a side API without a validation study would have allowed the thesis to claim a control that the software does not perform. The requirement is retained as future work in Chapter 9.

When the model service fails, the analysis service still stores the natural-language-inference view of the answer and sets the scoring model label to the local fallback. The model status field tells the recruiter that the frozen scorer did not cover every answered question. This is the operational form of the research rule against a fabricated DeBERTa score.

4.4 Detailed Design

The model service request contains three strings: the question, the reference answer, and the candidate answer. The runner rejects an empty question, an empty reference, or an empty answer. It builds the premise as the question plus the reference answer, obtains natural-language-inference probabilities for that premise against the candidate answer, computes the eight lexical features, and runs the DeBERTa

encoder. The fusion network takes the encoder's first-token vector, a masked mean-pooled vector, and the extra features. Its widths are 768 and then 384, with layer normalisation, GELU, and dropout. The classification head has five outputs, the regression head has one output, and the ordinal head has four outputs. The returned object contains the score on a 0 to 100 scale, the rating name, the confidence of the selected class, the full class probability map, and the model identifier.

The analysis service treats an explicit recruiter reference as source "provided". Otherwise it builds a short job-context reference from matching skills and question terms and sets the source to "job-context". Concept checks call the local inference model with a hypothesis that the answer demonstrates a named concept. A concept is marked contradicted only when the contradiction probability is at least 0.6. Otherwise it is covered when the inference label or the lexical overlap supports it, and not mentioned when neither support is present. This threshold reduces contradiction warnings that come from an uncertain inference.

Speech metrics are computed from the answer text and the timestamps of the question and the answer turns. Filler patterns include "um", "uh", "erm", "hmm", "you know", "I mean", "kind of", "sort of", "basically", "actually", and a restricted pattern for "like". The metric object reports total words, meaningful words, filler count, filler percentage, the filler breakdown, response time in seconds, and answer duration in seconds. None of these fields is an input to the DeBERTa predict call.

Review policy is conservative. An item needs review when it is unanswered, when the model rating is Wrong, Poor, or Average, or when the fallback classification itself requests review. Good and Excellent model ratings are counted as strong answers in the summary. The overall score shown in the summary is the mean of the per-answer scores for answered questions, rounded to the nearest integer. The recruiter interface follows the same separation. The interview room shows the live session and the transcript. The analysis panel and the result pages show the summary and the per-question evidence. Match score and attrition risk, when present elsewhere on the platform, stay on their own screens or fields. The interview page does not overwrite them.

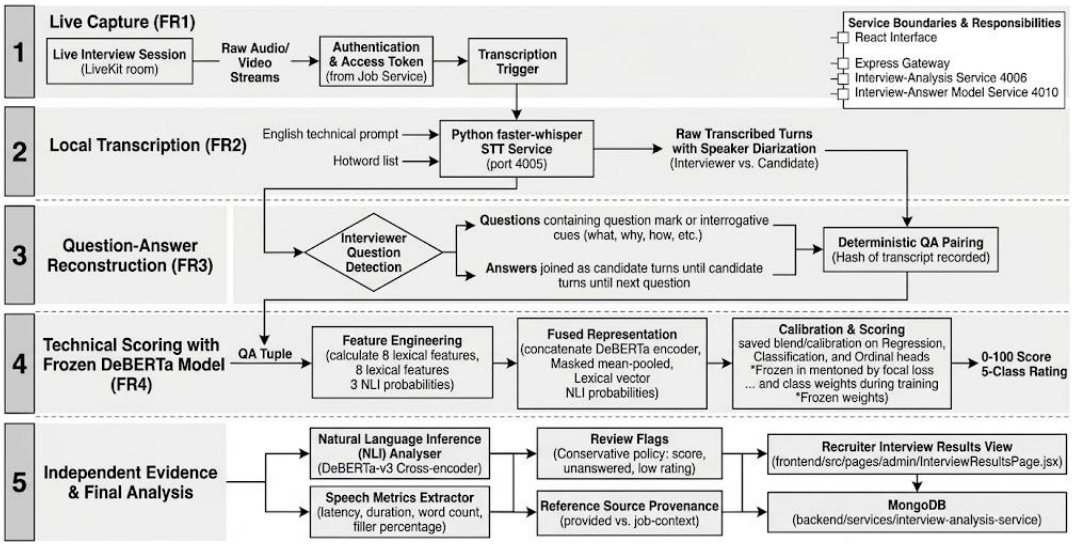


Figure 2 : Five-Stage Processing Pipeline / Workflow Diagram

4.5 Standards and Professional Practice

The design follows a documented service boundary for each model, an explicit request contract, health endpoints, and local loading of weights. Secrets for the LiveKit token remain in the job service configuration rather than in the browser source. The scoring service is stateless apart from the loaded weights, which allows the analysis service to retry or to record failure without corrupting the model. Logging and error responses use the platform's request-context pattern so that a failed predict call can be traced. Professional assessment practice is reflected in the rubric, the blinded rating procedure, and the decision to publish unfavourable pilot statistics. A quadratic weighted kappa of 0.259 is reported because that is the measured value. The product copy and this thesis describe the module as decision support for that reason.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The component is implemented in the shared Lanka Talent Insights repository. The web client is React 18 with Vite. Gateway, job, and interview-analysis services are Node.js and Express. The speech service and the answer-model service are Python FastAPI applications. Model code uses PyTorch and Hugging Face Transformers. Transcription uses faster-whisper. Container images are defined for deployment with Docker Compose. MongoDB persists interview analysis documents. Table 5.1 summarises the stack used by this component.

Table 7 : Technology Stack

Layer	Technology	Use in This Component
Interview User Interface	React, Vite, LiveKit Client	Provides the interview room, live transcript panel, and analysis panel.
Edge	Express API Gateway	Handles authenticated interview and analysis routes.
Realtime Media	LiveKit	Provides private audio and video rooms using access tokens.
Speech	Python, FastAPI, faster-whisper	Performs local English speech transcription using a technical prompt and domain-specific hotwords.
Analysis Orchestration	Node.js, Express	Manages question–answer pairing, NLI evidence, speech metrics, and score assembly.
Technical Scorer	Python, FastAPI, PyTorch, DeBERTa-v3	Performs frozen V5 multi-task interview-answer scoring.
Inference Features	Local DeBERTa NLI Cross-Encoder	Produces entailment, neutral, and contradiction probabilities.
Persistence	MongoDB	Stores the analysis results associated with each interview.
Packaging	Docker Compose	Provides a repeatable runtime environment.

5.2 Component Implementation

The speech service loads a Whisper model from the local model directory at startup and exposes transcription of an uploaded audio file. The decoder receives an English technical-interview prompt and a hotword string covering programming languages, data structures, databases, cloud services, and software-engineering terms. Voice activity detection and a beam size are configurable. If the model directory is missing, the service stays up and reports the load error on the health response, so that the rest of the platform can show a failed dependency instead of hanging.

The answer-model service loads the V5 configuration, the tokenizer, the DeBERTa state dictionary, and the natural-language-inference model during startup. Prediction runs under a lock so that concurrent requests do not interleave a single loaded network unsafely. The response is the calibrated score, rating, confidence, class probabilities, and model identifier described in Chapter 4. The service returns a client error when any of the three required strings is empty.

The analysis service implements pairing, reference planning, concept labelling, speech metrics, and the call to the model client. It hashes the source transcript so that a stored analysis can be related to the transcript version that produced it. The saved document includes the per-question block and the summary counts defined in Section 4.4. Authorised users can edit or delete a transcript turn before they request analysis, which is the practical control for repairing a recognition mistake.

Key implementation artefacts in the R26-IT-082 repository include:

- backend/services/speech-to-text-service/src/server.py (faster-whisper /transcribe)
- backend/services/interview-answer-model-service/src/model_runner.py (AdvancedASAGModel, lexical_features, predict)
- backend/services/interview-answer-model-service/src/app.py (POST /predict on port 4010)
- backend/services/interview-analysis-service/src/services/analysisService.js (runInterviewAnalysis)

- backend/services/interview-analysis-service/src/services/transcriptService.js (pairTranscript)
- backend/services/interview-analysis-service/src/services/scoringService.js (referenceFor, analyzeAnswer)
- backend/services/interview-analysis-service/src/services/metricsService.js (calculateSpeechMetrics)
- backend/services/interview-analysis-service/src/model/nliModel.js (classifyPair)
- backend/services/job-service/src/services/livekitService.js (room token)
- frontend/src/pages/interviews/LiveKitInterviewRoomPage.jsx
- frontend/src/pages/admin/InterviewResultsPage.jsx and InterviewResultDetailPage.jsx
- experiments/experiment_3_interview_human_validation/ (methodology, annotations, summary_metrics.json)

The browser room joins LiveKit with a token from the job service. The transcript panel can listen live and can fall back to batch Whisper transcription. The analysis panel requests analysis for the current interview and renders the stored result. Admin result pages list completed analyses and open one interview at a time.

5.3 Integration with the Other Components

Integration is by contract, not by a shared score. The interview-analysis service reads interview context from the platform, including the transcript and the job record produced by the vacancy and candidate services. Job skills and description text are inputs to question-relevance and to the fallback reference. They are not converted into the CV match percentage. The attrition service is not called during interview scoring, and the interview score is not written into the attrition feature vector by this component.

The gateway is the external entry used by the browser. Internal model calls use the service URLs configured for the analysis service, with the answer-model service as the scoring dependency. A recruiter can therefore schedule an interview in the shared

calendar, conduct it in the shared room, and open an analysis that belongs only to this module. The other members' screens continue to show their own evidence.

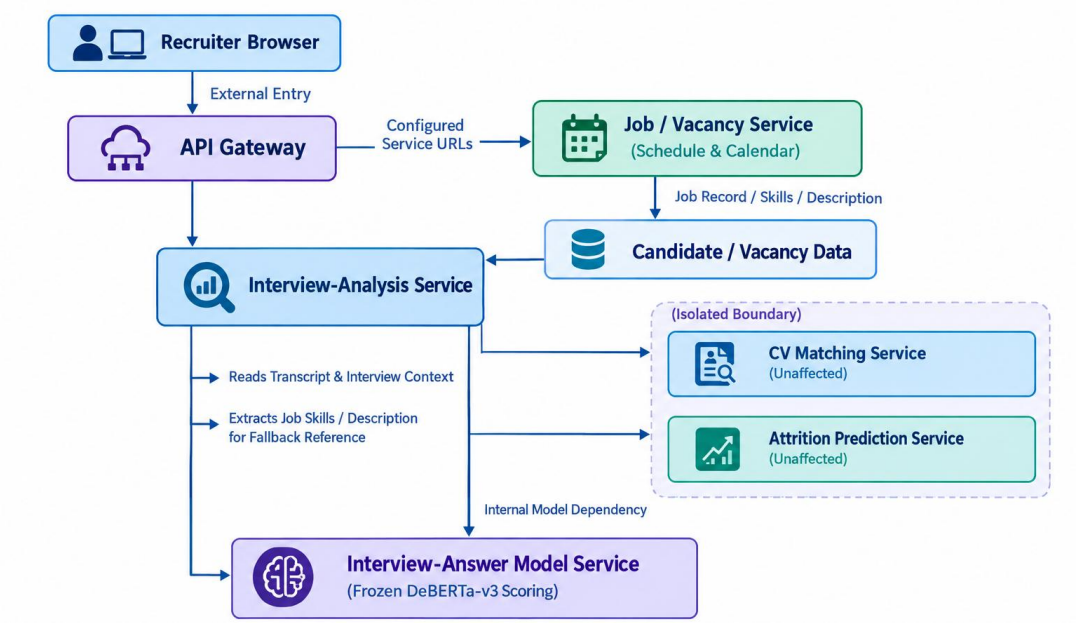


Figure 3 : Component integration and data flow across platform services

Figure 3 illustrates how the Interview-Analysis Service integrates with the API Gateway, Job/Vacancy Service, and Interview-Answer Model Service to evaluate candidate responses using the frozen DeBERTa-v3 model. It also shows that the existing CV Matching and Attrition Prediction services remain unaffected, ensuring modularity and minimal disruption to the platform.

5.4 Verification and Software Testing

Automated checks cover the deterministic parts of the analysis path. One test confirms that an interviewer question is paired with the following candidate turn and that the answer text is preserved. A second test confirms filler counting and response time: for a question at a known timestamp, an answer starting three seconds later, and the text "Um, I, like, build React components.", the metric function returns a response time of three seconds, two filler tokens, and six words. A third test confirms that a question about React, asked against a job that lists React, produces a job-context reference plan containing React when the recruiter has not supplied an

answer. These checks lock the pairing, speech, and reference-source behaviour that the scoring model depends on.

Table 5.2 records the verification that follows from those tests and from the service contracts. Items that were exercised by automated tests are marked as tested. Items that are specified by the service behaviour and still require a full end-to-end run against loaded model weights are marked as specified, so that a demonstration is not reported as an automated test result.

Table 8 : Implemented Verification Checks

Check	Expected Behavior	Evidence
Question Pairing	The candidate turn following an interviewer question is assigned as the answer.	Automated test
Unpaired Interviewer Text	Only interrogative interviewer turns open a question–answer pair.	Pairing rule in the transcript service
Filler and Latency Metrics	Filler-word counts and response times are calculated using transcript timestamps.	Automated test
Missing Reference	The reference source is derived from the job context, and the identified concepts may include matched skills.	Automated test
Empty Scoring Input	The model service rejects requests with a missing question, reference answer, or candidate answer.	Input validation in the model runner
Model Service Unavailable	The analysis service records the error and uses a scoring label that does not identify the result as a DeBERTa score.	Analysis orchestration branch
Health Checks	Speech and model services report their health status and whether model weights have loaded successfully.	Health endpoints

The human-validation script provides a separate verification of the research claim. It compares expert-assigned scores with the frozen model outputs for the ten evaluation items and generates the agreement table presented in Chapter 6. This process evaluates the model's performance without modifying or updating its weights.

5.5 Deployment and Operation

Local development starts the frontend, the gateway, and the model services as separate processes. Container deployment uses Docker Compose so that the speech model, the answer model, and the analysis service can be restarted independently. Weights are large and are kept in the service model directories. They are loaded once at process start. Replacing a model artefact requires a controlled restart of that service. It does not require a change to the React screens, provided the response fields remain the score, rating, confidence, probabilities, and model identifier.

Operation during an interview is sequential from the user's point of view. The room is opened, transcription is started, the conversation is stored as turns, and analysis is requested when the recruiter is ready. Analysis can be repeated after a transcript correction because the stored result records a hash of the transcript that was analysed. The recruiter should treat a job-context reference, a needs-review flag, or a model-unavailable status as a reason to read the answer, not as a finished hiring recommendation.

6 RESULTS AND VALIDATION

6.1 Research Results

RQ1, public short-answer test. On the 1,705-item held-out set the calibrated multi-task model achieved a quadratic weighted kappa of 0.758 and a Pearson correlation of 0.795. These figures indicate that the model learned both ordinal class structure and a continuous score that moves with the reference grades on leakage-cleaned educational answers. They answer RQ1 for that data regime. They are not interview-pilot figures, and the pilot error values are not copied into this table.

Table 9 : Held-Out Automatic Short-Answer Grading Performance

Metric	Value
Held-out Examples	1,705
Quadratic Weighted Kappa (QWK)	0.758
Pearson Correlation	0.795

Table 10 : Human Inter-Rater Reliability on the Ten-Item Pilot

Metric	Value
Evaluators	3
Items	10
Mean Pairwise Pearson Correlation (\(r\))	0.908
Mean Pairwise Spearman Correlation (\(\rho\))	0.883
Fleiss' Kappa	0.520
Mean Pairwise Quadratic Weighted Kappa (QWK)	0.535

Table 11 : Agreement with Human Consensus on the Ten-Item Pilot

Method	Pearson \(r\)	Spearman \(\rho\)	QWK	MAE	RMSE	Exact Match	Macro-F1
Proposed DeBERTa-v3 (Calibrated)	0.429 (p = 0.216)	0.587	0.259	9.06	11.58	60.0%	37.1%
Sentence-BERT (all-MiniLM-L6-v2)	-0.072 (p = 0.844)	0.006	-0.129	7.55	11.21	30.0%	15.4%
BERTScore (F1)	0.258 (p = 0.471)	0.447	0.000	7.19	8.83	10.0%	9.1%
ROUGE-L	0.562 (p = 0.091)	0.581	0.041	47.25	48.09	0.0%	0.0%

6.2 Product and Technical Results

The implemented path produces a stored analysis document for an interview. For each paired question the document can contain the question text, the joined candidate answer, the reference text, the reference source, relevance and coverage evidence, concept statuses, speech metrics, the model score and rating when the model responded, the model identifier, and a review flag. The summary aggregates those rows. Speech indicators available on each answered row are listed in Table 6.4. They are present so that a recruiter can see delivery. They are absent from the feature vector that produces the technical score.

Table 12 : Speech indicators reported beside the technical score

Indicator	Definition in the Implementation
Response Latency	The time in seconds between the question timestamp and the first candidate turn of the answer.
Answer Duration	The time in seconds between the first and last candidate turns combined into the answer.
Word Count	The number of tokens in the normalised answer text.
Meaningful-Word Count	The number of tokens remaining after the text utility removes non-content tokens.
Filler-Word Count	The number of occurrences of configured filler-word patterns and

Indicator	Definition in the Implementation
and Percentage	their percentage relative to the total word count.

Model health is a product result as well as a research safeguard. The scoring service and the speech service report load failure instead of pretending that a model is ready. The analysis summary reports how many answered questions received a model score and how many did not. A partial status is possible when some questions were scored and others were not.

6.3 User or Stakeholder Validation

The existing repository does not report a separate quantitative user study or stakeholder survey. Therefore, the validation presented in this thesis is **technical and functional** in nature rather than a controlled recruiter pilot or a statistically powered stakeholder evaluation. The available validation evidence is derived from the implemented system behaviour, service-level testing, and the internal consistency of the prediction, explanation, and simulation components.

Technical validation was performed at three levels. First, the model layer was validated using leakage-cleaned held-out test data, where the production DeBERTa-v3 scorer achieved a Quadratic Weighted Kappa of 0.758 and a Pearson correlation of 0.795. Second, the service layer was validated through functional test cases covering prediction, missing-value handling, scenario updates, explainability, service-continuity fallback, and health/readiness behaviour. All reported functional test cases passed. Third, the integration layer was validated by exercising the end-to-end workflow from the recruiter interface through the API Gateway, Interview Orchestrator, DeBERTa service, Early Attrition service, and explainability service. The reported integration tests confirm that the services share a stable model-ready contract, execute model inference, generate explanations, and re-run the same prediction path for What-If scenarios without mutating the stored candidate snapshot.

The technical validation also confirms that the system separates continuous probability from thresholded decision and from the percentage-scaled risk score shown to the recruiter. This separation is verified through the API responses, which

expose probability, threshold metadata, predicted class, risk level, and contributing factors as distinct outputs. The explainability service was validated to ensure that it does not recalculate the prediction or introduce a second scoring system; the mathematical attribution remains the source of feature contribution, while the contextual narrative only translates those model-derived drivers into recruiter-readable language.

No controlled user study, usability assessment, or stakeholder acceptance evaluation was conducted within the current scope. A recruiter pilot and stakeholder validation study are identified as future commercialization and validation steps. Therefore, the validation presented in this thesis should be interpreted as **technical validation of the implemented system**, not as empirical evidence of user acceptance or organizational impact.

6.4 Integration Results

The interview module is integrated to the extent that the shared workspace can schedule a room, attach a transcript, and request analysis from the interview-analysis service, which in turn calls the model service. Job context from the shared vacancy record is sufficient for a fallback reference, and the automated test in Chapter 5 shows that this fallback is labelled. The CV-matching service and the attrition service continue to expose their own results. No integration test in this component rewrites a match score from an interview rating. That absence is the intended integration result.

6.5 Summary of Findings

The public-data test supports RQ1. The model's held-out quadratic weighted kappa of 0.758 and Pearson correlation of 0.795 show useful graded structure on educational short answers. The pilot supports RQ2. Experts agree strongly on continuous totals and moderately on five classes. RQ3 receives a narrow answer. On this ten-item set the calibrated model has the best quadratic weighted kappa and the best exact class agreement among the four methods, and it does not have the best absolute error or a significant Pearson correlation. RQ4 is met in the design and the

implementation by local inference, named reference provenance, separation of speech evidence from the technical score, exclusion of demographic inputs, review flags, visible model status, and a human hiring decision. The findings justify continued development and a larger validation set. They do not justify autonomous rejection or selection of candidates.

7 DISCUSSION

7.1 Interpretation of Findings

The two experiments answer different questions, and the gap between them is informative. A quadratic weighted kappa of 0.758 on typed public answers shows that the training task was learned. A quadratic weighted kappa of 0.259 against interview experts shows that transfer into this pilot is weak. The drop is consistent with a change of domain, a change from typed text to interview-style answers, very small sample size, and a human label distribution that cannot stabilise a five-class kappa. It is not evidence that the held-out number was misreported, and the held-out number is not evidence that interview scoring is already reliable.

Human raters illustrate the same difficulty. A Pearson correlation of 0.908 means the experts ordered and scaled the answers similarly. A Fleiss' kappa of 0.520 means the five named bands still attracted disagreement. Some of that disagreement is the loss of information when a 100-point total crosses a boundary such as 74 versus 75. The product therefore shows both the continuous score and the class. A recruiter who sees 73 and Average, or 76 and Good, can treat the boundary as soft.

The baseline table warns against choosing a method by one column. BERTScore and Sentence-BERT can sit closer to the human total in absolute error on these ten items and still assign the wrong class most of the time. ROUGE-L can correlate in rank and remain far from the human numeric scale. The operational reason for preferring the calibrated model, if one method must be shown as the technical score, is that its class decisions matched the human class on 6 of 10 items and its ordinal kappa was the only positive value clearly above the overlap baselines. That reason is provisional until the item count reaches the planned range. Speech metrics are interpreted as context. A long pause or a high filler percentage may tell a recruiter something about delivery. It does not subtract points inside the model. This avoids punishing a concise correct answer, and it avoids rewarding a long incorrect answer. The human rubric still gives communication clarity 10 points, so the human total is not a pure technical-content score. Part of the remaining model-human gap may come from that

extra human dimension. The study did not partial out the communication marks, and the thesis does not claim that it did.

7.2 Comparison with Literature Baselines and Existing Solutions

The public-data result sits in the line of transformer short-answer graders that outperform surface matching on educational items [1]-[4], [11], [12]. The architecture extends those graders by adding an explicit lexical vector and a frozen inference vector before the three task heads, which is the response to the contradiction problem described by the natural-language-inference literature [6], [7]. The pilot then shows the limit of that extension: extra features and calibration improved class agreement relative to untuned similarity baselines, and they did not reach human-human ordinal agreement. Compared with interview platforms that emphasise conversation or facial analysis [13], [16], [18], [19], this system is narrower and more auditable. It will not claim an emotion or a personality score. Compared with a video store inside an applicant tracking system, it adds a question-level technical score and a concept list. Compared with a keyword match between the answer and the job description, it requires a reference and can mark a contradiction. The price of that design is dependence on reference quality. A job-context sentence assembled from skill names is a weaker premise than an expert paragraph, and weak premises should be read as weak evidence.

7.3 Research Contribution

The research contribution is a leakage-safe interview-scoring method and an independent human test of that method. The method binds local transcription and question pairing to a frozen multi-task DeBERTa-v3 scorer that uses contextual representations, eight lexical features, and three inference probabilities, with saved calibration and an ordered five-class output. The human test uses a six-dimension rubric and grade bands that are not the model's thresholds, three blinded raters, and three published baselines on the same items. The contribution includes the negative and the inconclusive numbers. Reporting a non-significant Pearson correlation and a quadratic weighted kappa below the pre-registered target is part of the result, because it identifies what a larger study must improve.

7.4 Product Contribution

The product contribution is the interview path inside the group platform: a private room, local transcription, persisted turns, an analysis record, and a recruiter view of score, class, confidence, reference source, concepts, contradictions, unanswered questions, and speech indicators. The contribution is also the boundary with the other members. Interview performance is an additional evidence channel. It is not a replacement for skill extraction, job match, or attrition risk, and it is not an automatic offer.

7.5 Limitations and Threats to Validity

The pilot has ten items. Correlation intervals at that size are wide, the Pearson p-value of 0.216 does not support a claim of linear agreement, and a handful of items in one score band can move kappa sharply. The pre-registered plan asked for 60 to 100 items across frontend, backend, database, cloud, and system-design topics. That plan was not completed. Experts may have shared a similar background, which can inflate human-human correlation and still leave the model short of their consensus.

Public training answers are clean and typed. Speech-recognition mistakes are a threat that this thesis does not quantify. Job-context references are a threat to construct validity whenever a recruiter does not write a reference. English-only software questions are a threat to external validity for Sinhala or Tamil conversation, for behavioural questions, and for design questions with several acceptable solutions. The human communication marks are a threat to interpretation if they are treated as if the model had been asked to predict them. Model and serving equivalence remains a threat until every reported score is regenerated from the production predict endpoint on the final annotation file. No subgroup fairness audit by accent, gender, or first language was performed. Because the scorer sees text, unfairness can still enter through recognition errors and through references that privilege one phrasing of a correct idea. The thesis does not claim demographic parity.

7.6 Implications

For research, the next valid step is a larger blinded set, confidence intervals, and a paired comparison of manual versus automatic transcripts. For practice, a recruiter can use a Good or Excellent rating as a prompt to read a strong answer, and a Wrong, Poor, Average, unanswered, or model-unavailable status as a prompt to read carefully. The score is a sorting and explanation aid. Offers, rejections, and feedback to candidates remain human acts. For the group platform, the implication is to keep the four analytical outputs visible and separate so that a weak interview does not erase a strong CV match, and a strong interview does not hide an attrition concern raised by another component.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

8.2 Market and Competitive Position

8.3 Business or Sustainability Model

8.4 Scalability Intellectual Property and Compliance

8.5 Contribution to the Sustainable Development Goals

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the AI-based interview-evaluation component of the AI-Powered Intelligent Talent Acquisition System. The component transcribes a live English technical interview locally, reconstructs question-answer pairs, scores answered questions with a frozen multi-task DeBERTa-v3 model, and shows natural-language-inference evidence and speech indicators without folding delivery into the technical mark. On public short-answer data the model reached a quadratic weighted kappa of 0.758 and a Pearson correlation of 0.795. On a tri-blind pilot of ten software-engineering items, three experts agreed strongly with each other on continuous scores, and the model's agreement with their consensus was preliminary: exact class agreement of 60%, quadratic weighted kappa of 0.259, and a non-significant Pearson correlation of 0.429. The system is a recruiter aid. It is not an autonomous hiring system.

9.2 Achievement of Objectives

Local capture, transcription, question pairing, unanswered detection, frozen scoring, calibrated 0 to 100 output, five-class ratings, reference labelling, concept and contradiction evidence, separate speech metrics, and visible model failure were specified and implemented. The human comparison and the three baselines were completed for a ten-item pilot and were reported with the metrics required to see both the strengths and the shortfalls. Integration into the shared interview workflow was implemented so that analysis sits beside, and does not replace, matching and attrition. The certification-authenticity objective from the Topic Assessment Form was not achieved in the evaluated software and is not claimed. The pre-registered sample of 60 to 100 interview items was not achieved. The objectives that concern a trustworthy product boundary were achieved. The objective of demonstrated high agreement with human interviewers was only partially addressed and remains open.

9.3 Individual Research and Product Contributions

The individual research contribution is the reference-conditioned DeBERTa scoring design, the separation of technical and speech evidence, and the non-circular human validation with an honest pilot result. The individual product contribution is the interview-analysis path in the group system, from the LiveKit room and local Whisper transcription to the stored recruiter report. The contribution stops at the boundary of the other members' components.

On the research side, the reference-conditioned scoring design treats each candidate answer as a tuple of question, answer, and reference answer, and combines a DeBERTa-v3 encoder representation with eight lexical coverage features and three frozen natural-language-inference probabilities. The design is reference-conditioned because the same answer can be correct for one question and irrelevant for another, so the score depends on the question and the reference rather than on the answer alone. The separation of technical and speech evidence is the second research contribution. Response latency, answer duration, word count, filler-word frequency, and filler-word percentage are computed and reported separately from the technical score, so that hesitation, speaking speed, or accent cannot reduce a technically correct answer. The third research contribution is the non-circular human validation. The model weights, calibration values, and class-decision rule were frozen before the human study, and the human scores were not used for training, calibration, or model selection. The expert rubric is independent of the model's learned thresholds, which prevents the experts from being asked to reproduce the model's own grading rule. The pilot result is reported honestly, including the finding that model-human agreement was below human-human agreement and that the ten-item sample is too small to support broad reliability claims.

On the product side, the interview-analysis path begins at the LiveKit interview room where the recruiter and candidate communicate, and where candidate audio is captured. The audio is transcribed locally using a Whisper model, and the transcript is reconstructed into question-answer pairs. Each pair is passed to the DeBERTa scorer, and the NLI analyzer produces concept coverage and contradiction evidence alongside the technical score. Speech metrics are extracted from the transcript and

timestamps. The results are then stored and presented in the recruiter report, where the technical score, predicted class, model confidence, concept coverage, contradiction warnings, unanswered-question status, communication indicators, and model-availability status are displayed as separate evidence dimensions. If the scorer is unavailable, the system records that failure explicitly and does not fabricate a score.

The contribution stops at the boundary of the other members' components. Resume parsing, job-candidate matching, and attrition prediction are implemented by the other group members. The interview-evaluation component consumes the candidate and job context those components provide, and contributes its own output to the shared recruiter dashboard, but it does not modify their models, their feature sets, or their decision logic. This boundary keeps the three evidence dimensions distinct and allows each component to be developed, evaluated, and replaced independently.

9.4 Future Directions

The immediate research task is to expand the blinded set from 10 items to at least 60, and preferably 100, balanced across frontend, backend, database, cloud, and system-design questions. The same items should be scored from corrected transcripts and from automatic transcripts so that speech-recognition damage can be measured by word error rate and by the change in the technical score. Confidence intervals, a larger-sample test of correlation, and an intraclass correlation for the raters should be added. Stronger baselines can remain Sentence-BERT, BERTScore, and ROUGE-L, provided every score is mapped with a declared rule onto the 0 to 100 scale before absolute error is compared. A fairness review should test whether accents and code-switching among English, Sinhala, and Tamil technical speech change recognition and scores. Multilingual transcription is a product requirement for many Sri Lankan interviews and was not solved here. Recruiter studies should measure whether the concept list and the review flag change decisions, time, and trust. Certificate validation, as named in the Topic Assessment Form, should be a separate design with an authentic and a forged document set, and it should not be mixed into the technical-answer score until that study exists. Finally, the production predict

endpoint should be the only scoring path used in the camera-ready evaluation, so that the numbers in the thesis and the numbers on the recruiter screen refer to the same function.

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VI. GLOSSARY

Automatic Short-Answer Grading: The process of scoring a free-text response by comparing it with a reference answer or a learned grading pattern.

Calibration: The saved linear mapping applied to the blended model output before presenting it as a score from 0 to 100.

Exact Class Agreement: The percentage of items for which the predicted quality class matches the human consensus class.

Fleiss' Kappa: A statistical measure used to evaluate agreement among more than two raters on categorical labels, adjusted for agreement occurring by chance.

Job-Context Reference: A fallback reference answer created from the job vacancy and interview question when the recruiter has not provided an expected answer. It is clearly labelled and is not treated as an expert reference.

Natural Language Inference (NLI): The process of classifying a candidate's text as entailed by, contradictory to, or neutral with respect to a given premise.

Quadratic Weighted Kappa: A statistical measure of agreement that penalises classification errors according to the square of the distance between the assigned classes.

Reference Provenance: The recorded source of the expected answer, indicating whether it was provided by the recruiter or constructed from the job context.

Tri-Blind Rating: An evaluation procedure in which each expert independently scores a response without seeing the model output, other experts' scores, or candidate demographic data.

VII. APPENDICES

Appendix A Research Instruments or Extended Results

The human evaluation instrument consists of the six-dimension rubric presented in Table 3.2. Three experts independently evaluate each response, and their scores are averaged to obtain the consensus score. The continuous consensus scores are mapped according to the procedure described in Table 3.3. The public-data split is summarised in Table 3.1. The pilot agreement statistics presented in Tables 6.2 and 6.3 are obtained from the project's human-validation evaluation using the ten-item expert file. The held-out Quadratic Weighted Kappa and Pearson correlation values presented in Table 6.1 are the public-data results reported for the frozen scoring model. These values are not recalculated using the ten interview items. The speech-metric instrument identifies the filler-word patterns implemented in the analysis service. These include "you know", "I mean", "kind of", "sort of", "um", "uh", "erm", "hmm", "basically", "actually", and a comma-bound pattern for "like". Response time and duration are calculated using transcript timestamps. These measurements are stored in the corresponding question records and are not included in the scoring request.

Appendix B Academic Integrity and AI Use Disclosure

enerative Artificial Intelligence (AI) was used as a writing aid to organise this individual thesis according to the supplied chapter structure, align the content with the Topic Assessment Form, the individual research proposal, the DeBERTa research paper, and the implemented interview services, and improve grammatical accuracy and consistency. Numerical results were obtained from the author's research paper and the project's human-validation summary. No numerical results were fabricated to improve the perceived performance of the proposed system. In particular, the pilot evaluation results, including cases in which baseline models outperform the proposed model on certain error metrics, are reported as observed. The author retains responsibility for the research design, implementation, data, citations, analyses, interpretations, and conclusions presented in this thesis. Before submission, the

author must verify all citations, tables, supervisor names, and final pagination in Microsoft Word. Signature fields have intentionally been left blank for completion where required. No generative AI system was authorised to modify model weights, fabricate interview transcripts, or replace the independent assessments of human experts. Research findings concerning attrition, salary, and workforce surveys that belong to another group member are outside the scope of this individual thesis and are therefore excluded from the interview-evaluation findings.